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Digital Communications

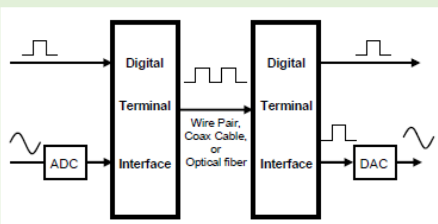
Digital Communications

Digital Communications covers digital transmission and digital radio. It gradually replaced the traditional analog systems that use conventional AM, FM and PM.

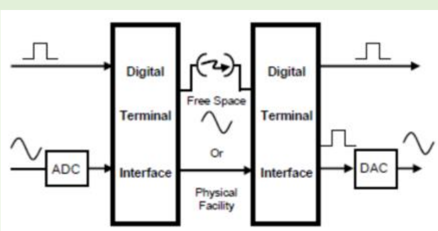
Digital Transmission vs Digital Modulation/Radio

Digital Transmission is a transmittal of digital pulses between two points in a communication system while **Digital Radio** is a transmittal of digitally modulated analog carrier between two points in a communication system.

Digital Transmission Block Diagram



Digital Modulation Block Diagram



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Digital Modulation

Digital Radio or Digital Modulation is the property that distinguishes a digital radio system from a conventional AM, FM, or PM radio system is that in a digital radio system the modulating and demodulated signals are digital pulses rather than analog waveforms.

Amplitude Shift Keying (ASK)

In ASK, the amplitude of the analog carrier is varied by the digital modulating pulse. The modulating signal is a binary pulse stream that varies between two discrete voltage levels rather than continuously changing analog signal as in conventional AM.

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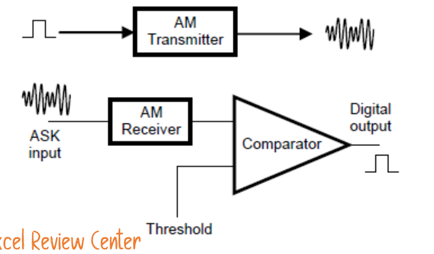

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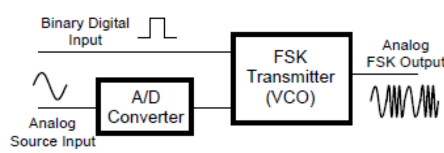
Frequency Shift Keying (FSK)

In FSK, the frequency f the analog carrier is varied by the digital modulating pulse. In its simplest form, two frequencies are generated, one corresponding to a binary zero (space) and the other a binary one (mark).

FSK is reliable in the presence of noise since each signal has only two possible states. FSK is also used in high-frequency radio systems to transmit teletype information.

FSK Transmitter



FSK Waveform

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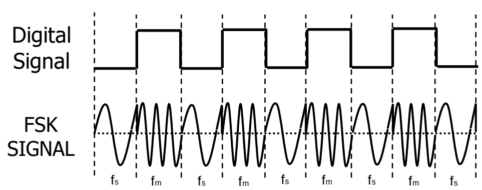

Important Notes:

- The center or carrier frequency is shifted by a binary input data.
- The FSK output shifts between two frequencies: a mark or logic 1 frequency, a space or logic 0 frequency.
- There is a change in the output frequency each time the logic conditions of the binary input signal changes.
- Bit rate and Baud rate are equal.

General expression of an FSK Signal

$$v(t) = V_C \cos\left[\left(\omega_c \pm \frac{f_m(t)\Delta\omega}{2}\right)t\right]$$

where:
$v(t)$ = binary FSK waveform (V)
V_c = peak unmodulated carrier amplitude (V)
ω_c = radian carrier frequency
$f_m(t)$ = binary digital modulating signal frequency

Modulation Index (m_f)

$$m_f = \frac{\text{frequency deviation}}{\text{modulating frequency}} = \frac{\delta}{f_a}$$

$$m_f = \frac{\left|\frac{f_m - f_s}{2}\right|}{\frac{f_b}{2}} = \frac{|f_m - f_s|}{2}$$

where:
δ = frequency deviation (Hz)
f_m = mark frequency (Hz)
f_s = space frequency (Hz)
$\left|\frac{f_m - f_s}{2}\right|$ = peak frequency deviation
f_b = input bit rate
$\frac{f_b}{2}$ = fundamental frequency of the binary input signal

Modulating Frequency or Fundamental Frequency (f_a)

$$f_a = \frac{f_b}{2}$$

Frequency Deviation (δ)

$$\delta = \frac{f_m - f_s}{2} \quad (\text{if } f_m > f_s)$$

$$\delta = \frac{f_s - f_m}{2} \quad (\text{if } f_s > f_m)$$

Minimum Bandwidth Required (f_N)

$$f_N = 2nf_a$$

where:
n = number of significant sidebands (dependent on the modulation index similar to conventional FM)
f_a = modulating frequency (Hz)

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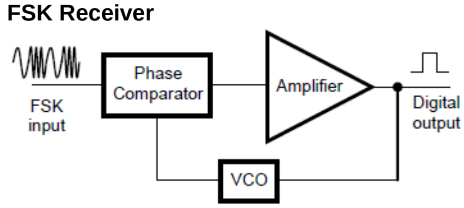
Note:
That the minimum bandwidth required propagating a signal is called the **minimum Nyquist bandwidth (f_N)**. When modulation is used and the double-sided output spectrum is generated, the minimum bandwidth is called the **minimum double-sided Nyquist bandwidth or the minimum IF bandwidth**.

FSK Receiver



Minimum Shift Keying (MSK)

Also known as Continuous – Phase Frequency Shift Keying (CPFSK). It is essentially a binary FSK except that the mark and space frequencies are synchronized with the input binary bit rate (f_b).